Confluence Project
Genotyping and Imputation QC Process

This document describes the Confluence data processing steps for samples genotyped at NCI/CGR and Cambridge University using the Confluence GSAv3 and GDA arrays. The Confluence data includes newly genotyped GSA and GDA samples. All GDA data were generated by NCI/CGR. GSA data were generated from two centers, Cambridge and NCI.

GSA and GDA data summary

1. Data processing workflow

The data processing steps are summarized in the workflow in Figure 1.

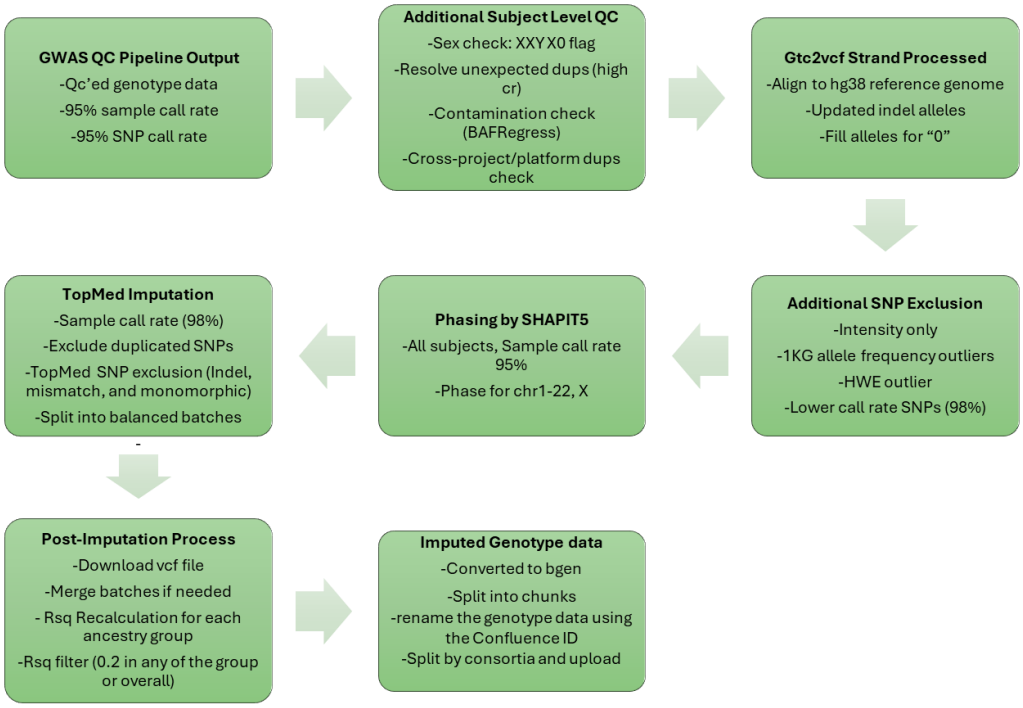

Figure 1 Confluence GSA and GDA data processing workflow

2. Quality control (QC) filtering and genotype data freeze

For Confluence GSA dataset, NCI, LAGENO and CLINICAL samples were genotyped at CGR. The samples from India were genotyped at an external institute and the raw data were transferred to CGR and QC'ed together with NCI, LAGENO and CLINICAL samples.
For Confluence GDA dataset, all samples were genotyped at CGR. All GSA and GDA samples were QC'ed using CGR's in-house GWAS QC pipeline v1.7.0rc1 (https://github.com/NCI-CGR/GwasQcPipeline).

2.1 QC analysis for individual dataset
A set of standard QCs was first applied at the sample level, then followed by the subject level check. Here is a brief description of the QC steps:

-Self-cluster EGT file generation for GSA and GDA:
To generate the EGT cluster file for the GSA array we used Genome Studio and Illumina's GenTrain 3.0 algorithm on a set of 60,577 samples. We included 356 samples from the 1000 Genomes Project and selected the best 250 samples for each Confluence study by call rate. We included 52% of the 7,259 LAGENO consortium samples, and 2819 male samples for Y chromosome clustering. We aimed to include a diverse set of samples to maximize chances of having heterozygotes for the rarer variants. Seven Illumina scanners were used in genotyping. The proportion of total samples scanned from each scanner that were included in cluster set ranged from 33% to 62%.

The GDA self-clustered egt was generated at CGR and followed similar processes as the GSA egt. We created a Genome Studio Project and used Illumina's GenTrain 3.0 algorithm on a set of 6239 samples available at the time, including 269 male samples for Y chromosome clustering. We aimed to include a diverse set of samples to maximize chances of having heterozygotes for the rarer variants. Four Illumina scanners were used in the genotyping, with approximately equal proportions of samples from each scanner included in the cluster set.

-Completion Rate Filter: Iterative 80% and 95% sample- and variant-level call rate filters were applied. Samples below call rate filters were excluded.

-Contamination Check: Sample contamination rate was examined on samples genotyped at CGR using the VerifyIDintensity tool (https://genome.sph.umich.edu/wiki/VerifyIDintensity). Samples that passed the completion rate filters or had a median raw intensity > 6000 (internal tests show this tool to be unreliable for low intensity data). Samples with predicted contamination > 20% were excluded. Samples with predicted contamination rate between 10%-20% were further reviewed using two additional tools: BAFRegress (https://genome.sph.umich.edu/wiki/BAFRegress)and baf_auto generated from CGR's in-house Contamination Prediction Pipeline. Samples contaminated after review were excluded at the subject level.

-Genotype Concordance Check-Expected Duplicates: Replicate samples are included in every project to be used to assess overall project quality. PLINK was used to perform all vs. all concordance check for all samples and the discordant expected duplicates were removed when the concordance rate was <95%. To convert the data from sample level to subject level,

the replicate with the highest sample completion rate was kept and the other replicates were removed.

-**Abnormal Heterozygosity**: The autosomal heterozygosity rates were calculated for all subjects using plink –het function. These are available in supporting_files/autosomal_het/.

-**Sex Check**: Sex was verified by comparing the reported sex with the observed sex based on X chromosome method-of-moments F coefficient ([observed hom. count] - [expected count]) / ([total observations] - [expected count]) using PLINK. F coefficient is expected to be close to 0.0 and 1.0 for females and males, respectively. The threshold is set to 0.5, the midpoint of these expected values. Samples that were found to have sex chromosome aneuploidies including XXY, X0 and mosaic XX were flagged, while true sex discordant (reported sex differed from genetic sex) subjects were excluded.

-**Genotype Concordance Check-Unexpected Duplicates**: Pairwise genotype concordance for all subjects was assessed to discover unexpected replicates. Subjects with genotype concordance greater than 95% at a set of LD-pruned SNPs were considered replicates. For each unexpected replicate pair, the higher call rate subject was kept and the lower call rate subject was excluded.

-**Relatedness Check**: Identity-by-descent (IBD) for all subject pairs were computed for close relationships (1st and 2nd degree) using PLINK.

-**Ancestry Inference**: Ancestry was estimated using the GRAF-pop software. GRAF-pop (https://github.com/ncbi/dbsnp/tree/graf-pop-test/Frequency Project/graf-pop) calculates genetic distances from each sample to reference populations and estimates subject ancestry and ancestral proportions based on these distances.

-**PCA analysis**: PCA analysis was performed using smartpca (https://github.com/chrchang/eigensoft/blob/master/POPGEN/smartpca.info). The top 10 PCs were calculated and PCA plots were generated per genetic ancestry group to evaluate case-control balance.

2.2 Cross dataset duplicates check

Pairwise genotype concordance for all subjects from GDA and GSA and the majority of previous genotyping was assessed to discover cross-dataset unexpected replicates (different Confluence subject ID with concordant genotype – greater than 95% concordance).

3. Additional SNP Quality control (QC) filtering, Phasing, and Batching for GSA

3.1 Align to hg38 reference genome

The genotype data in the Illumina top strand orientation were aligned to hg38 reference genome (GCA_000001405.15_GRCh38_full_analysis_set.fna) using the bcftools gtc2vcf plugin (https://github.com/freeseek/gtc2vcf).

3.2 Further SNP exclusion before phasing

We removed intensity-only probes, SNPs with call rate below 98%, those with discordant frequencies compared to the 1000 Genomes project and Hardy-Weinberg Equilibrium (HWE) test outliers. HWE outliers were calculated for control samples using PLINK --hwe function with pvalue threshold E-12. For GSA data, allele frequencies were compared with the 1000 Genomes reference EUR and EAS populations, and discordant SNPs were excluded (the comparison method is described in Michailidou et al. (https://pubmed.ncbi.nlm.nih.gov/29059683/)); for GDA data, HWE outliers were identified for AFR population. A combined list of SNP exclusions was generated from the union of the Cambridge and NCI lists.

3.3 Phasing by SHAPEIT

Phasing was performed using SHAPEIT (v5.1.1; https://mathgen.stats.ox.ac.uk/genetics_software/shapeit/shapeit.html).

3.4 Further low call rate subjects and SNP exclusions after phasing before submitting for imputation

Subjects below 98% call rate were further removed before imputation. The phased genotype data was uploaded to the TOPMed imputation server for additional QC check by TOPMed. The indels, mismatched SNPs, duplicated SNPs and monomorphic SNPs identified by the TOPMed QC were further removed from the phased genotype data.

3.5 Batching for GSA data

Due to the limitations of the TOPMed imputation server, the submitted data was split into batches of up to 25,000 samples.

3.6 Imputation with TOPMed Imputation Server

Batches of GSA data and GDA data were submitted to the TOPMed imputation server with the settings below:

atalyst.nhlbi.nih.gov/#!run/imputationserver%402.0.0-beta3

NIH National Heart, Lung, and Blood Institute | BioData CATALYST TOPMed Imputation Server

Home Run Jobs About Help Contact sali

Genotype Imputation (Minimac4) 2.0.0-beta3

This is the new Imputation Server Pipeline using Minimac4. Documentation can be found here.

If your input data is **GRCh37/hg19** please ensure chromosomes are encoded without prefix (e.g. **20**).
If your input data is **GRCh38/hg38** please ensure chromosomes are encoded with prefix 'chr' (e.g. **chr20**).

There is a limit of three concurrent jobs per person. The TOPMed imputation server is a free resource, and these limits allow us to provide service to a wide audience. We reserve the right to terminate users who violate this policy. https://topmedimpute.readthedocs.io/en/latest/

Run

Name: Confluence GSA batch 1

Reference Panel (Details): TOPMed r3

Input Files (VCF):

- subjects-chr1.phased.filtered_new.batch_1.vcf.gz
- subjects-chr2.phased.filtered_new.batch_1.vcf.gz
- subjects-chr3.phased.filtered_new.batch_1.vcf.gz
- subjects-chr4.phased.filtered_new.batch_1.vcf.gz
- subjects-chr5.phased.filtered_new.batch_1.vcf.gz
- subjects-chr6.phased.filtered_new.batch_1.vcf.gz
- subjects-chr7.phased.filtered_new.batch_1.vcf.gz
- subjects-chr8.phased.filtered_new.batch_1.vcf.gz

Select Files

Multiple files can be selected by using the ctrl / cmd or shift keys.

Array Build: GRCh38/hg38

Please note that the final SNP coordinates always match the reference build.

rsq Filter: off

Phasing: No phasing

Population: vs. TOPMed Panel

Mode: Quality Control & Imputation

☐ AES 256 encryption

Imputation Server encrypts all zip files by default. Please note that AES encryption does not work with standard unzip programs. Use 7z instead.

☑ Generate Meta-imputation file

NOTE: Submitted data is only used for imputation, and is deleted after job completion.

☑ I will not attempt to re-identify or contact research participants.

☑ I will report any inadvertent data release, security breach or other data management incident of which I become aware.

Submit Job

4. Post-imputation processing

After downloading the imputed data from the TOPMed imputation server, the three batches of GSA imputed data were merged using bcftools. During the merging process, bcftools calculates the average values of the R^2, allele frequency (AF), and minor allele frequency (MAF) for each variant.

4.1 Recalculate R2 by GrafAnc ancestry group using SNPTEST

For GSA imputed data, R2 values were recalculated for subjects of EUR, SAS and AMR ancestry based on GrafAnc ancestry assignment. For GDA imputed data, R2 values were recalculated for subjects of AFR ancestry based on GrafAnc ancestry assignment. R2 values were recalculated using SNPTEST v2.5.6 (https://www.chg.ox.ac.uk/~gav/snptest/). SNPs that have both recalculated R2=1 and MAF=0 were filtered out.

4.2 Imputed data SNP filtering

For GSA imputed data, the imputed SNPs were retained for downstream analysis if their R2 values are greater than 0.2 in any ancestry-recalculated R2 (EUR, EAS, SAS or AMR) or in the overall imputed R2. The retained SNPs for the samples processed at CGR also were merged with the retained SNPs of the samples processed at Cambridge following the same SNP filtering process.

For GDA imputed data, the imputed SNPs were retained for downstream analysis if their R2 values are greater than 0.2 in AFR ancestry-recalculated R2 or in the overall imputed R2.

4.3 Split chromosome into chunks and convert to bgen format

The filtered imputed data were split into chunks for parallel computing using bcftools v1.20. The vcf chunk files were then converted into bgen format using qctools v2.2.0 (https://www.chg.ox.ac.uk/~gav/qctool_v2/).

4.4 Subset subjects by consortium

The filtered imputed data were also subset based on subjects in each consortium.